

Корневая фс

1) Собрать корневую файловую систему состоящую из одного процесса `init`, который выводит строку "Hello" и засыпает на 15 секунд. Проверить работоспособность также в qemu.

```
> touch init.c
> nano init.c
> cat init.c
#include <stdio.h>
#include <unistd.h>

int main(void) {
    printf("Hello\n");
    fflush(stdout);
    sleep(15);
    return 0;
}
```

написали программку

```
> arm-linux-gnueabi-gcc -static -Os -o init init.c
> ls
init  init.c  rootfs
~/init
```

статически скомпилировали под arm

```
> echo init | cpio -o -H newc | gzip > initramfs.cpio.gz
1144 блока
~/E/Eltex_tasks/18/rootfs develop *1 !12 ?22 > █
```

создали initramfs с init

```
> cp ~/Eltex/Eltex_tasks/18_ROOT_FILE_SYSTEM/rootfs/initramfs.cpio.gz
> ls
initramfs.cpio.gz  linux-5.10.186  linux-5.10.186.tar.xz  rootfs
> cd ~/arm-qemu
```

запустили в qemu:

```
qemu-system-arm -M vexpress-a9 -kernel linux-5.10.186/arch/arm/boot/zImage -dtb
linux-5.10.186/arch/arm/boot/dts/vexpress-v2p-ca9.dtb -initrd initramfs.cpio.gz -append
"console=ttyAMA0" -nographic
```

```

ALSA device list:
  #0: ARM AC'97 Interface PL041 rev0 at 0x10004000, irq 32
Freeing unused kernel memory: 1024K
Run /init as init process
Hello
input: ImEXPS/2 Generic Explorer Mouse as /devices/platform/bus@400000000/bus@400000000:moth
Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000
CPU: 0 PID: 1 Comm: init Not tainted 5.10.186 #1
Hardware name: ARM-Versatile Express
[<8010f244>] (unwind_backtrace) from [<8010afa8>] (show_stack+0x10/0x14)
[<8010afa8>] (show_stack) from [<8081adec>] (dump_stack+0x94/0xa8)
[<8081adec>] (dump_stack) from [<80816e24>] (panic+0x124/0x32c)
[<80816e24>] (panic) from [<80124ba0>] (do_exit+0x984/0x9cc)
[<80124ba0>] (do_exit) from [<80125b9c>] (do_group_exit+0x3c/0xb8)
[<80125b9c>] (do_group_exit) from [<80125c28>] (__wake_up_parent+0x0/0x18)
---[ end Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000 ]---

```

BUSYBOX

2) Собрать корневую файловую систему на основе busybox и запустить в qemu с init процессом в виде командного интерпретатора /bin/ash.

Скачала busybox-1.36.1 и плату vexpress-v2p-ca9.dtb

создали дефолтный конфиг: ARCH=arm make defconfig

```

* klogd should not be used together with syslog to kernel printk buffer
*
Use the klogctl() interface (FEATURE_KLOGD_KLOGCTL) [Y/n/?] (NEW) y
logger (6.3 kb) (LOGGER) [Y/n/?] (NEW) y
logread (4.8 kb) (LOGREAD) [Y/n/?] (NEW) y
Double buffering (FEATURE_LOGREAD_REDUCED_LOCKING) [Y/n/?] (NEW) y
syslogd (13 kb) (SYSLOGD) [Y/n/?] (NEW) y
Rotate message files (FEATURE_ROTATE_LOGFILE) [Y/n/?] (NEW) y
Remote Log support (FEATURE_REMOTE_LOG) [Y/n/?] (NEW) y
Support -D (drop dups) option (FEATURE_SYSLOGD_DUP) [Y/n/?] (NEW) y
Support syslog.conf (FEATURE_SYSLOGD_CFG) [Y/n/?] (NEW) y
Include milliseconds in timestamps (FEATURE_SYSLOGD_PRECISE_TIMESTAMPS) [N/y/?] (NEW) n
Read buffer size in bytes (FEATURE_SYSLOGD_READ_BUFFER_SIZE) [256] (NEW) 256
Circular Buffer support (FEATURE_IPC_SYSLOG) [Y/n/?] (NEW) y
Circular buffer size in Kbytes (minimum 4KB) (FEATURE_IPC_SYSLOG_BUFFER_SIZE) [16] (NEW) 16
Linux kernel printk buffer support (FEATURE_KMSG_SYSLOG) [Y/n/?] (NEW) y

```

последние логи команды

открыли ARCH=arm make menuconfig, задали префикс и static build

```

*** End of configuration.
*** Execute 'make' to build the project or try 'make help'.

/home/alyssa/Eltex/Eltex_tasks/18_ROOT_FILE_SYSTEM/busybox-1.36.1
arm-linux-gnueabihf-: команда не найдена
make[1]: arm-linux-gnueabihf-: Нет такого файла или каталога
make[1]: arm-linux-gnueabihf-: Нет такого файла или каталога

~/E/Eltex_tasks/18/busybox-1.36.1 develop *1 !12 ?23 >

```

ARCH=arm make defconfig
make -j \$(nproc)
Всё собралось:

make install - делаем каталог для формирования фс

```
You will probably need to make your busybox binary
setuid root to ensure all configured applets will
work properly.
-----

> file ./busybox
./busybox: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically
a1]=6844eac4dd1f3c043e5e7704dc6a94384699b063, for GNU/Linux 3.2.0, stripped

~/E/Eltex_tasks/18/busybox-1.36.1 develop *1 !12 ?23 >
```

видно, что arm

```
> cd _install
> ls
bin linuxrc sbin usr

~/E/Eltex_tasks/18/b/_install develop *1 !12 ?23 >
```

```
> rm -rf linuxrc
> find . | cpio -o -H newc | gzip > initramfs.cpio.gz
2238 блоков

~/E/Eltex_tasks/18/b/_install develop *1 !12 ?23 >
```

заменяем:

```
> ls
init initramfs.cpio.gz
> rm -rf initramfs.cpio.gz

~/E/Eltex_tasks/18/rootfs develop *1 !12 ?23 >
```

```
> cp initramfs.cpio.gz ../../rootfs/
> ls
bin initramfs.cpio.gz sbin usr

~/E/Eltex_tasks/18/b/_install develop *1 !12 ?23 >
```

wget <https://cdn.kernel.org/pub/linux/kernel/v5.x/linux-5.10.186.tar.xz>

tar xf linux-5.10.186.tar.xz

cd linux-5.10.186

подготовили конфиг: make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi-hf-
vexpress_defconfig

собираем ядро: `make -j$(nproc) ARCH=arm CROSS_COMPILE=arm-linux-gnueabi-hf-zImage`

по итогу в каталоге:

```
> ls
busybox-1.36.1  linux-5.10.186  rootfs  zImage
init.c         linux-5.10.186.tar.xz  vexpress-v2p-ca9.dtb

~/E/Eltex_tasks/18_ROOT_FILE_SYSTEM develop *1 !12 ?23 > █
```

`QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -dtb vexpress-v2p-ca9.dtb -initrd initramfs.cpio.gz -append "console=ttyAMA0 rdinit=/bin/ash" -nographic`

Запустилось:

```
aaci-pl041 10004000.aaci: FIFO 512 entries
NET: Registered protocol family 17
9pnet: Installing 9P2000 support
oprofile: using arm/armv7-ca9
Registering SWP/SWPB emulation handler
drm-clcd-pl111 10020000.clcd: DVI muxed to daughterboard 1 (core tile) CLCD
drm-clcd-pl111 10020000.clcd: initializing Versatile Express PL111
drm-clcd-pl111 10020000.clcd: DVI muxed to daughterboard 1 (core tile) CLCD
drm-clcd-pl111 10020000.clcd: initializing Versatile Express PL111
ALSA device list:
  #0: ARM AC'97 Interface PL041 rev0 at 0x10004000, irq 41
Freeing unused kernel memory: 1024K
Run /bin/ash as init process
/bin/ash: can't access tty; job control turned off
~ #
~ #
~ #
~ #
~ #
~ #
~ # ls
bin          initramfs.cpio.gz  sbin
dev          root              usr
~ # █
```

создали директорию proc и sys, смонтировали их

```
ps: can't open '/proc': No such file or directory
~ # mkdir proc
~ # mkdir sys
~ # mount -t proc proc proc
~ # ls /proc
1      28      68      filesystems  self
10     3       7       fs          slabinfo
11     30     8       interrupts  softirqs
12     31     9       iomem       stat
13     32    asound     ioports     swaps
14     33    buddyinfo irq         sys
15     36    bus       kallsyms    sysrq-trigger
16     4     cgroups    kmsg        sysvipc
17     46    cmdline    kpagecount  thread-self
18     47    config.gz  kpageflags  timer_list
19     48    consoles  loadavg     tty
2      49     cpu        locks       uptime
20     50    cpuinfo    meminfo     version
21     51    crypto     misc         vmallocinfo
22     52    device-tree modules      vmstat
23     53    devices    mounts       zoneinfo
24     58    diskstats  mtd
25     6     driver     net
26     61    execdomains pagetypeinfo
27     62    fb         partitions
~ #
```

```
27     62    fb         partitions
~ # ps
PID  USER      TIME  COMMAND
1  0          0:00  /bin/ash
2  0          0:00  [kthreadd]
3  0          0:00  [rcu_gp]
4  0          0:00  [rcu_par_gp]
6  0          0:00  [kworker/0:0H-ev]
7  0          0:00  [kworker/u8:0-ev]
8  0          0:00  [mm_percpu_wq]
9  0          0:00  [ksoftirqd/0]
10  0         0:00  [rcu_sched]
11  0         0:00  [migration/0]
12  0         0:00  [cpuhp/0]
13  0         0:00  [cpuhp/1]
14  0         0:00  [migration/1]
15  0         0:00  [ksoftirqd/1]
16  0         0:00  [kworker/1:0]
17  0         0:00  [kworker/1:0H]
18  0         0:00  [cpuhp/2]
19  0         0:00  [migration/2]
20  0         0:00  [ksoftirqd/2]
21  0         0:00  [kworker/2:0]
22  0         0:00  [kworker/2:0H]
23  0         0:00  [cpuhp/3]
24  0         0:00  [migration/3]
25  0         0:00  [ksoftirqd/3]
```

ps работает